# Challenge Assignment

Aprendizagem Computacional/Machine Learning, 2025
DEI, FCT, University of Coimbra

## Context ang Goals

The objective of the challenge is to encourage students (groups of two) promoting creativity and knowledge in the field of machine learning, with following guidelines.

1. **Objective**: The objective of the challenge is to encourage students to analyze a real-world problem that can be solved using machine learning and then construct simple models to tackle it.

2. **Case study**: A clear description of the proposed case study should be provided, including a brief introduction to the problem, a dataset description, and a detailed explanation of the used methodology.

3. **Data Source**: This synthetic dataset comprises quantitative morphological measurements of individual dry bean specimens drawn from multiple cultivated varieties. Each observation corresponds to a single bean and includes features, such as length, width, projected area, perimeter, roundness, and compactness. These variables capture both size-related and shape-related characteristics and exhibit the natural intercorrelations typical of biological measurement data. The objective is to predict if the bean is "**White Haricot Bean**" or not.
   https://bit.ly/binary_beans_english

4. **Model Construction**: To tackle the problem outlined in this assignment, students must construct at least two simple machine learning models, Decision Trees and K-Nearest Neighbors are suggested, but clustering approaches or Logistic Regression can also be used. The models should be constructed using Python programming language and should be able to make predictions on new unobserved data.

5. **Evaluation Metrics**: The models should be evaluated using appropriate evaluation metrics, which may vary depending on the problem being solved. Students should choose evaluation metrics that are appropriate for their problem and explain their choice. Visual aids can also be used.

6. **Documentation**: Students should document the entire process, from data collection to model construction, and evaluation. They should include details of the methodology used and the results obtained, along with any challenges faced and how they overcame them.

7. **Submission**: The final submission should include a brief report, the code used to construct the model, the documentation, and any other relevant information.

8. **Criteria**: The submissions will be evaluated based on the data preprocessing, choice of models, experimental setup, hyperparameter tuning, the performance of the model, the choice of evaluation metrics, the clarity and soundness of the documentation, and the discussion.

## Deliverables and deadlines

1. Challenge submission deadline: November 17th, 13h00

    a. Submit a .pdf file into *Inforestudante* with your report including all details described above.
    b. Submit a zip file with all your code including setup information (data, software, etc.)

2. Discussion is mandatory and will take place in the same day.